ECET 250E-Linear Circuit 1
Problemset 9-Chapter 10- Sinusoidal Steady State Analysis
Solution

Name: _____ Marks: _____

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1. **Problem 7.** Use nodal analysis to find **V** in the circuit of Fig. 10.56

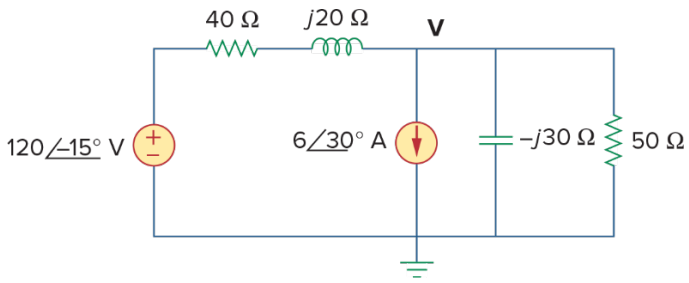


Figure 10.56 For **Prob. 10.7**.

Chapter 10, Solution 7.

At the main node,

$$\frac{120\angle -15^{o} - V}{40 + j20} = 6\angle 30^{o} + \frac{V}{-j30} + \frac{V}{50} \quad \longrightarrow \quad \frac{115.91 - j31.058}{40 + j20} - 5.196 - j3 =$$

$$V\left(\frac{1}{40 + j20} + \frac{j}{30} + \frac{1}{50}\right)$$

$$V = \frac{-3.1885 - j4.7805}{0.04 + j0.0233} = \underline{124.08\angle -154^{o}\ V}$$

2. **Problem 25.** Solve for i_o in Fig. 10.73 using mesh analysis.

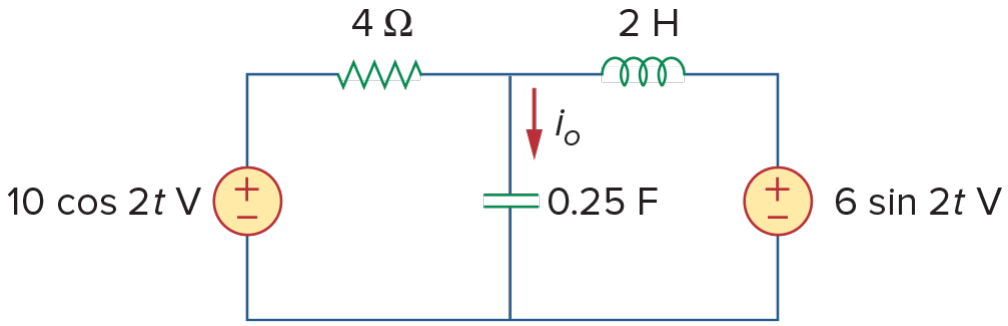


Figure 10.73 For **Prob. 10.25**.

Chapter 10, Solution 25.

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

$\omega = 2$

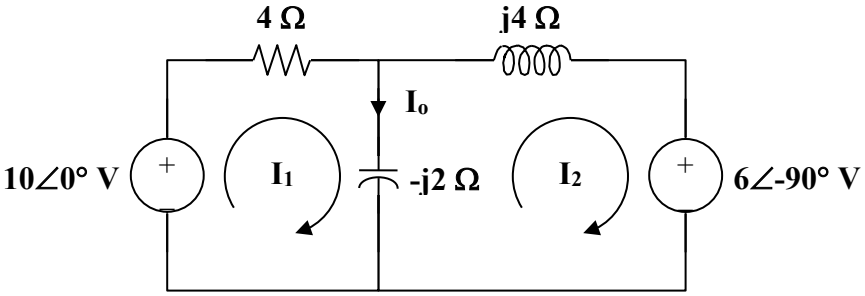
$$10\cos(2t) \longrightarrow 10\angle 0°$$

$$6\sin(2t) \longrightarrow 6\angle -90° = -j6$$

$$2\text{ H} \longrightarrow j\omega L = j4$$

$$0.25\text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(2)(1/4)} = -j2$$

The circuit is shown below.



For loop 1,

$$-10 + (4 - j2)\mathbf{I}_1 + j2\mathbf{I}_2 = 0$$

$$5 = (2 - j)\mathbf{I}_1 + j\mathbf{I}_2 \qquad (1)$$

For loop 2,

$$j2\mathbf{I}_1 + (j4 - j2)\mathbf{I}_2 + (-j6) = 0$$

$$\mathbf{I}_1 + \mathbf{I}_2 = 3 \qquad (2)$$

In matrix form (1) and (2) become

$$\begin{bmatrix} 2-j & j \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\Delta = 2(1-j), \qquad \Delta_1 = 5 - j3, \qquad \Delta_2 = 1 - j3$$

$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2 = \frac{\Delta_1 - \Delta_2}{\Delta} = \frac{4}{2(1-j)} = 1 + j = 1.4142\angle 45°$$

Therefore,

$$i_o(t) = \mathbf{1.4142\cos(2t + 45°)\ A}$$

3. **Problem 40.** Find i_o in the circuit shown in Fig. 10.85 using superposition.

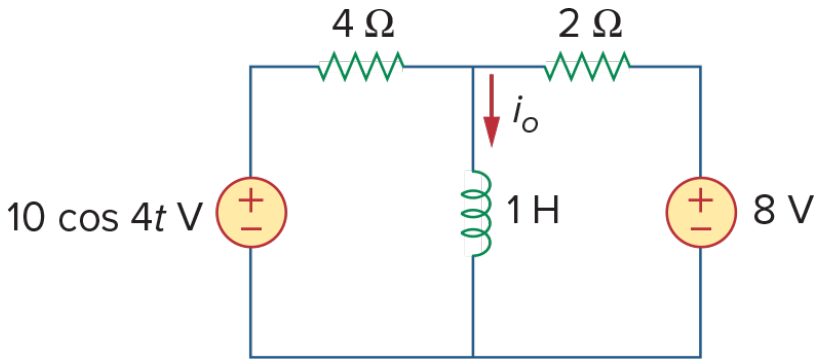
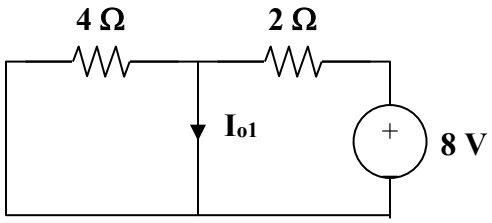


Figure 10.85 For **Prob. 10.40**.

Chapter 10, Solution 40.

Let $I_o = I_{o1} + I_{o2}$, where I_{o1} is due to the dc source and I_{o2} is due to the ac source. For I_{o1}, consider the circuit in Fig. (a).

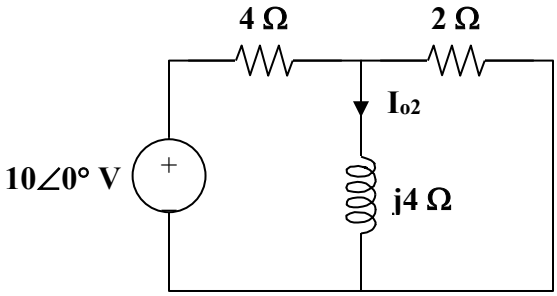


(a)

Clearly,

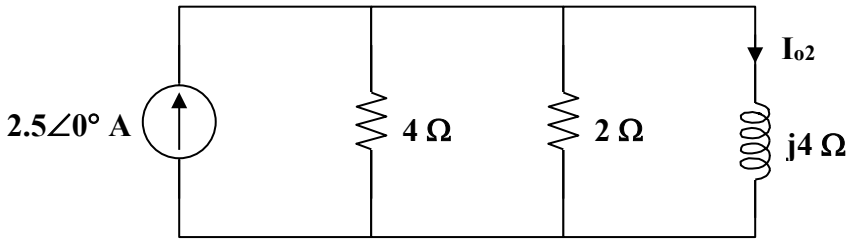
$$I_{o1} = 8/2 = 4 \text{ A}$$

For I_{o2}, consider the circuit in Fig. (b).



(b)

If we transform the voltage source, we have the circuit in Fig. (c), where $4 \,||\, 2 = 4/3\ \Omega$.



(c)

By the current division principle,

$$\mathbf{I}_{o2} = \frac{4/3}{4/3 + j4}(2.5\angle 0°)$$

$$\mathbf{I}_{o2} = 0.25 - j0.75 = 0.79\angle -71.56°$$

Thus, $$I_{o2} = 0.79\cos(4t - 71.56°)\ A$$

Therefore,

$$I_o = I_{o1} + I_{o2} = \mathbf{[4 + 0.79\cos(4t - 71.56°)]\ A}$$

4. **Problem 49.** Using source transformation, find *i* in the circuit of Fig. 10.94.

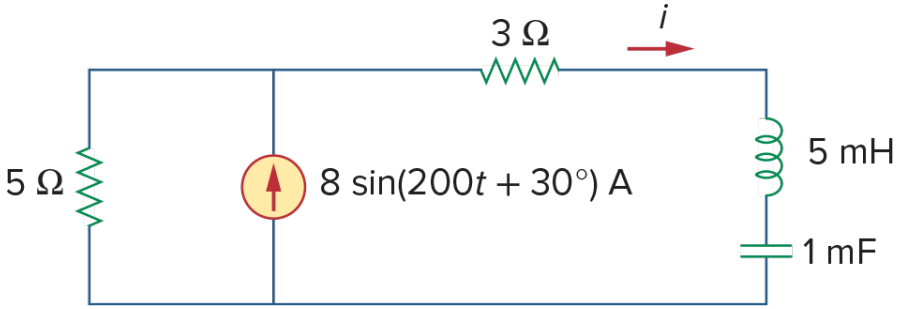


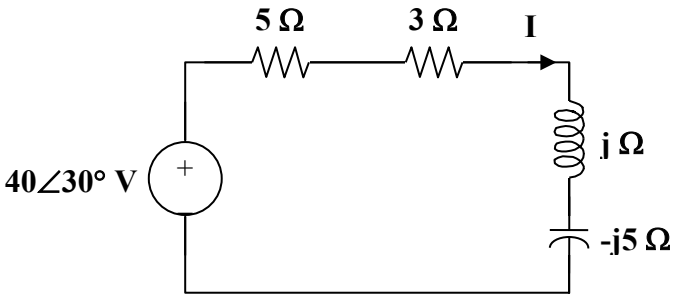
Figure 10.94 For **Prob. 10.49**.

Chapter 10, Solution 49. $8\sin(200t + 30°) \longrightarrow 8\angle 30°, \quad \omega = 200$

$$5\text{ mH} \longrightarrow j\omega L = j(200)(5\times 10^{-3}) = j$$

$$1\text{ mF} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(200)(1\times 10^{-3})} = -j5$$

After transforming the current source, the circuit becomes that shown in the figure below.



$$\mathbf{I} = \frac{40\angle 30^\circ}{5+3+\mathrm{j}-\mathrm{j}5} = \frac{40\angle 30^\circ}{8-\mathrm{j}4} = 4.472\angle 56.56^\circ$$

$$\mathrm{i} = \mathbf{[4.472\sin(200t+56.56^\circ)]\ A}$$

5. **Problem 55.** Find the Thevenin and Norton equivalent circuits at terminals *a-b* for each of the circuits in Fig. 10.98.

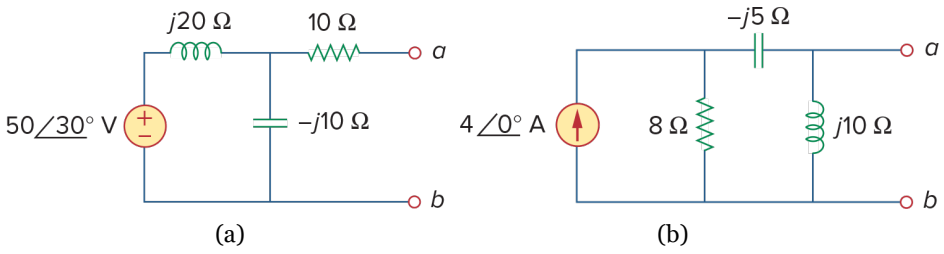
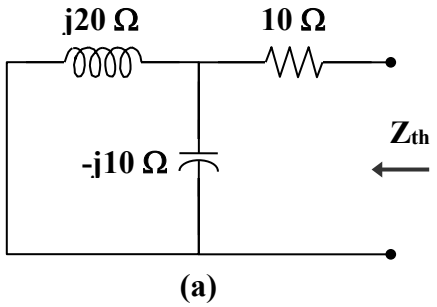


Figure 10.98 For **Prob. 10.55**.

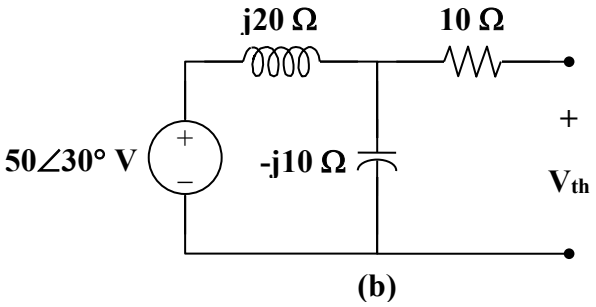
Chapter 10, Solution 55.



(a) To find $\mathbf{Z}_{\mathrm{th}}$, consider the circuit in Fig. (a).

$$\mathbf{Z}_N = \mathbf{Z}_{th} = 10 + j20 \,||\, (-j10) = 10 + \frac{(j20)(-j10)}{j20 - j10}$$

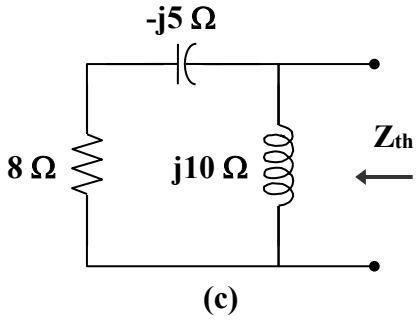
$$= 10 - j20 = \quad \mathbf{22.36\angle -63.43^\circ\ \Omega}$$



To find $\mathbf{V}_{th}$, consider the circuit in Fig. (b).

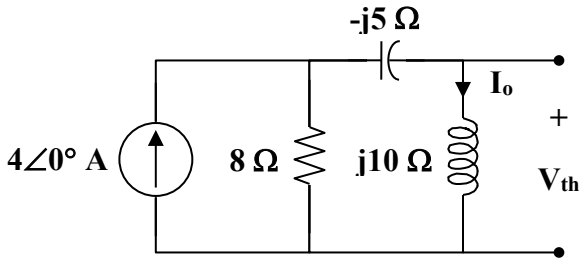
$$\mathbf{V}_{th} = \frac{-j10}{j20 - j10}(50\angle 30^\circ) = \quad \mathbf{50\angle -150^\circ\ V}$$

$$\mathbf{I}_N = \frac{\mathbf{V}_{th}}{\mathbf{Z}_{th}} = \frac{-50\angle 30^\circ}{22.36\angle -63.43^\circ} = \quad \mathbf{2.236\angle -86.6^\circ\ A}$$



(b) To find $\mathbf{Z}_{th}$, consider the circuit in Fig. (c).

$$\mathbf{Z}_N = \mathbf{Z}_{th} = j10 \,||\, (8 - j5) = \frac{(j10)(8 - j5)}{j10 + 8 - j5} = \mathbf{10\angle 26^\circ\ \Omega}$$



(d)

To obtain $\mathbf{V}_{th}$, consider the circuit in Fig. (d).

By current division,

$$\mathbf{I}_o = \frac{8}{8 + j10 - j5}(4\angle 0^\circ) = \frac{32}{8 + j5}$$

$$\mathbf{V}_{th} = j10\mathbf{I}_o = \frac{j320}{8 + j5} = \mathbf{33.92\angle 58^\circ\ V}$$

$$\mathbf{I}_N = \frac{\mathbf{V}_{th}}{\mathbf{Z}_{th}} = \frac{33.92\angle 58^\circ}{10\angle 26^\circ} = \mathbf{3.392\angle 32^\circ\ A}$$